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Introduction

In the realm of industrial instrumentation and electromagnetic compatibility (EMC), a Solid State Tesla Coil (SSTC) can be effectively utilized as a localized signal jammer by generating intense broadband electromagnetic interference (EMI). Unlike traditional jammers that target specific frequency bands using tuned RF oscillators, an operating SSTC produces high-voltage, high-frequency electric fields that naturally emit a wide spectrum of ambient radio-frequency noise. When this noise floor is raised high enough, it overwhelms the signal-to-noise ratio of nearby receiver circuits, effectively disrupting or “jamming” communication signals.

From an instrumentation and control perspective, generating this type of broadband interference is highly relevant for testing system robustness. By deploying a Tesla coil as a controlled EMI source, engineers can evaluate the shielding effectiveness of industrial sensors, verify the resilience of protective relay coordination, and ensure that critical load flow data acquisition systems remain stable and uncorrupted under extreme electromagnetic stress.

Components Used

Here is a breakdown of the components used in the Tesla Coil circuit, detailing their general functions and specific roles in generating broadband EMI.

1.1 Power Adapter (12V, 5A)

What it is: A power supply unit that converts AC mains power to direct current (DC).

Project Relevance: Serves as the primary energy source. The 5-ampere capacity provides the high current required by the transistor to drive the primary coil.



1.2 PVC Pipe and 30AWG Copper Wire

What it is: A non-conductive plastic form wrapped with very thin, enamel-insulated copper wire.

Project Relevance: Forms the **secondary coil**. The hundreds of turns act as a massive voltage step-up transformer, producing the high-voltage electric field.



Figure 2: PVC Pipe and 30AWG Copper Wire

1.3 BD139 BJT

What it is: An NPN bipolar junction transistor (BJT) designed for medium-power applications.

Project Relevance: Acts as the high-speed oscillator (switch). It rapidly pulses the DC current into the primary coil at the resonant frequency.

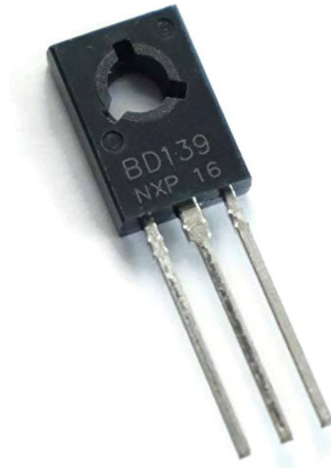


Figure 3: BD139 Transistor

1.4 2.2 kΩ Resistor

What it is: A passive electrical component that restricts the flow of electric current.

Project Relevance: Provides the necessary base bias current to “turn on” the transistor, kickstarting the self-oscillating feedback loop.

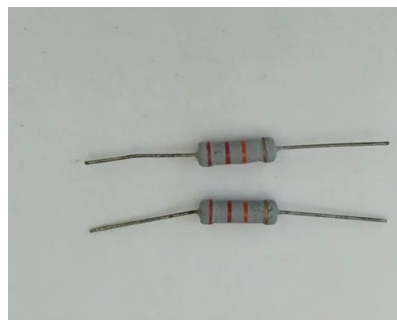


Figure 4: 2.2 kΩ Resistor

1.5 UF4007 Diode

What it is: An ultra-fast recovery rectifier diode.

Project Relevance: Acts as a protective flyback diode. It protects the sensitive base of the BD139 from destructive high-voltage reverse spikes.



Figure 5: UF4007 Diode

1.6 Multistranded Copper Wire (Primary Winding)

What it is: A highly conductive, flexible wire made of multiple smaller copper strands.

Project Relevance: Forms the **primary coil**. It carries the heavy, high-frequency current pulses from the transistor.



Figure 6: Primary Winding

1.7 Aluminium Foil (Top Load)

What it is: A thin, flexible sheet of conductive metal.

Project Relevance: Acts as the capacitive **top load**. It stores electrical charge and forms an LC resonant circuit, acting as the primary antenna that radiates the jamming frequencies.



Figure 7: Top Load

1.8 Heatsink

What it is: A thermally conductive metal component.

Project Relevance: Attached to the BD139 transistor to absorb and radiate the heat generated by rapid switching, preventing thermal runaway.

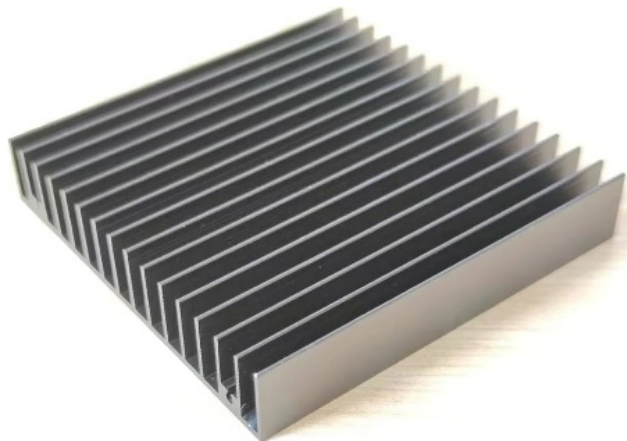


Figure 8: Heatsink

Working Principle

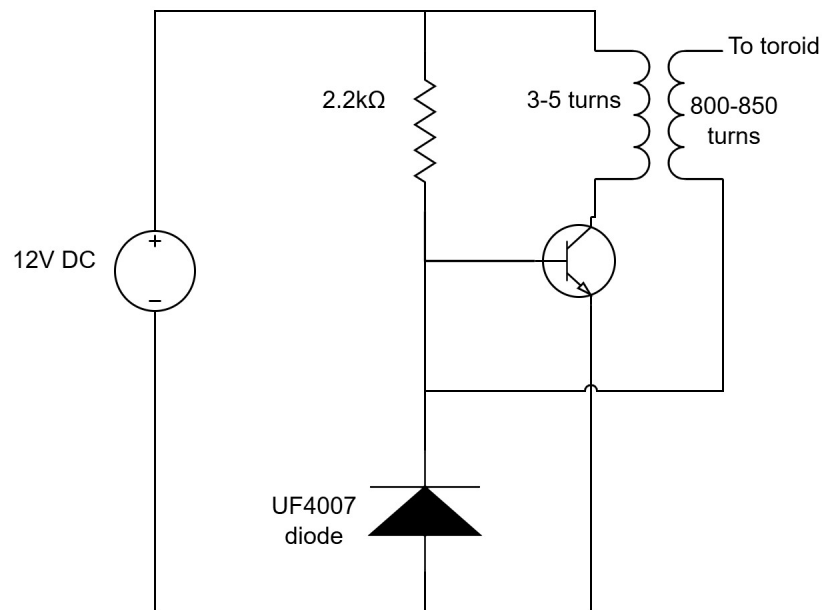


Fig: Circuit diagram

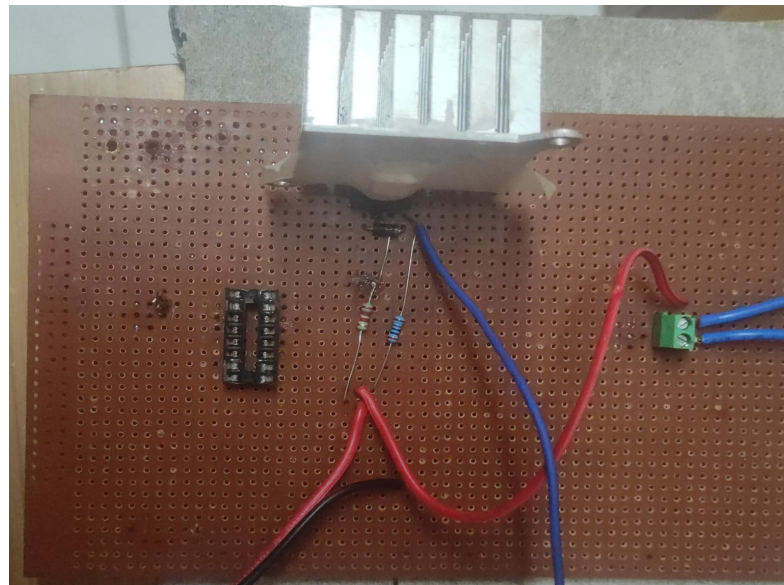


Figure 10: Final Project

The project utilizes a classic Slayer Exciter circuit topology, functioning as a highly efficient, self-resonating solid-state Tesla coil. The step-by-step operation is detailed below:

1. **Initial Biasing:** When the 12V DC power is connected, current travels through the 2.2 k Ω resistor directly to the Base of the BD139 transistor, biasing it into a conductive state.
2. **Primary Excitation:** With the transistor ON, a large surge of current flows through the thick primary winding into the Collector, out the Emitter, to ground. This generates a rapidly expanding magnetic field.
3. **Induction and Feedback:** The expanding magnetic field cuts across the secondary coil, inducing a voltage. The induced voltage at the bottom of the secondary coil swings sharply negative, neutralizing the positive base bias and snapping the transistor OFF. The UF4007 diode shunts any dangerous negative spikes to ground, protecting the transistor.
4. **The Flyback Effect:** As the transistor turns OFF, the primary magnetic field collapses rapidly. This induces a massive high-voltage spike in the secondary winding, which accumulates on the Aluminium foil top load.
5. **Oscillation Cycle:** Once the high voltage dissipates, the negative feedback disappears, and the 12V supply biases the base again. This rapid switching occurs at hundreds of kilohertz to megahertz, dictated by:

$$f = \frac{1}{2\pi\sqrt{LC}} \quad (1)$$

1.9 Mechanism of Signal Jamming

The circuit acts as a brute-force jammer by generating broadband Electromagnetic Interference (EMI). The violent switching of the BD139 creates a harmonic-rich “square-ish” wave. The aluminium foil top load acts as an antenna, forcefully radiating these high frequency fields. When a receiver is placed nearby, its receiving antenna picks up this chaotic RF noise, which swamps the legitimate signal. The signal-to-noise ratio (SNR) collapses, resulting in a jammed connection.

Applications of Signal Jammers

Using Tesla Coil

1. Military and Defense Applications

- Used in electronic warfare systems to disrupt enemy communication.
- Can interfere with radio-controlled devices and surveillance systems.
- Helps in protecting sensitive military zones from unauthorized wireless communication.

2. Security and Anti-Spy Systems

- Prevents unauthorized transmission of confidential information.
- Can block hidden wireless microphones or spy devices in secure rooms.
- Used in research on electromagnetic shielding and interference protection.

3. Research and Educational Purposes

- Demonstrates the effect of electromagnetic interference (EMI) on communication systems.
- Used in laboratories to study high-frequency electromagnetic wave behaviour.
- Helps students understand wireless communication vulnerabilities.

4. Testing of Electronic Equipment

- Used for EMI/EMC testing to check how electronic devices behave under strong electromagnetic fields.
- Helps improve shielding techniques in communication systems.

5. Protection Against Unauthorized Drones

- Experimental systems may interfere with radio signals used by drones.
- Studied for securing airports, military bases, and restricted areas.

6. Industrial and Scientific Experiments

- Used in experiments involving plasma generation and high-voltage RF phenomena.
- Helps analyze electromagnetic compatibility in industrial electronics

Conclusion

In conclusion, the “Signal Jammer using Tesla Coil” project successfully demonstrates the generation of broadband electromagnetic interference (EMI) through a highly efficient, solid-state Slayer Exciter circuit. By rapidly switching a low-voltage DC supply through a resonant transformer network, the system generates intense, localized high-frequency electric fields. These fields effectively raise the ambient RF noise floor, overwhelming the signal-to-noise ratio of nearby receivers and jamming local communication signals. From an industrial instrumentation perspective, this project serves as a powerful practical model for understanding the disruptive nature of EMI, highlighting the critical importance of proper shielding and electromagnetic compatibility (EMC) in the design of robust sensor networks and control systems.